COURSE NAME & CODE: ECA 0303 – Signals and systems

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Experiment 5.2

Step Response of the System

Aim:

To analyze the step response of a discrete-time system. The step response characterizes how the system responds to a step input, which is a common way to assess the transient and steady-state behavior of the system.

Program:

```
% Test Case: Step Response of a Discrete-Time System
% Define the length of the signal
N = 20; % Length of the signal

% Define the input signal (step function)
u_n = ones(1, N); % Step function

% Define the system parameters
b = [1];      % Numerator coefficients of the system (input)
a = [1, -0.5]; % Denominator coefficients of the system (output)

% Compute the system's response to the step input
y_n = filter(b, a, u_n); % System output for step input
% Define the time index for the signals
n = 0:N-1; % Time index
```

```
% Plot the step input and system response
clf; % Clear the current figure
% Plot step input
subplot(2,1,1);
bar(n, u_n, 'b');
xlabel('n');
ylabel('u[n]');
title('Step Input Signal');
% Plot system response to step input
subplot(2,1,2);
bar(n, y_n, 'r');
xlabel('n');
ylabel('y[n]');
title('Step Response of the System');
```

Output:

Result:

1. Plot of Step Input u[n]

o A stem plot showing the step input signal, which will be a constant value of 1 for all n.

2. Plot of Step Response y[n]:

o A stem plot showing the system's output y[n] in response to the step input signal. This plot will illustrate how the system evolves over time from the moment the step input is applied.

This test case will help you visualize and analyze the step response of a discrete-time system, providing insights into its transient and steady-state behavior.